

12

D

$$U = \frac{3}{2}NkT = \frac{1}{2}Nm(c_{rms})^2$$

$$\frac{T}{(c_{rms})^2} = \text{constant}$$

$$\left(\frac{T}{(c_{rms})^2} \right)_{initial} = \left(\frac{T}{(c_{rms})^2} \right)_{final}$$

$$\frac{330}{(c_{rms})^2} = \frac{T_{final}}{(120\% \times c_{rms})^2}$$

$$T_{final} = 475 \text{ K}$$

13

A

With damping, the amplitude will be lower for all frequencies due to loss in energy.